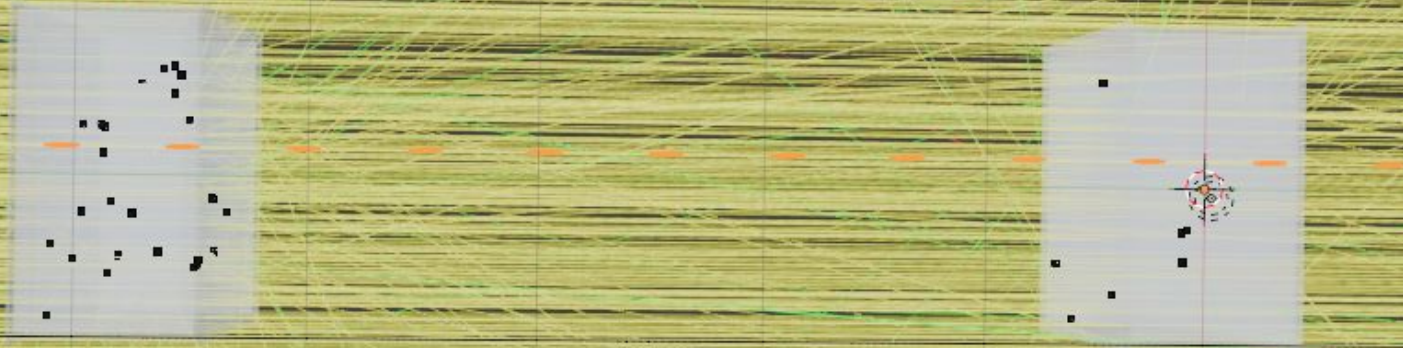




Simulating Single-Scatter Neutron Spectrometers and the nDURE Instrument for Detecting High Energy Albedo Neutrons



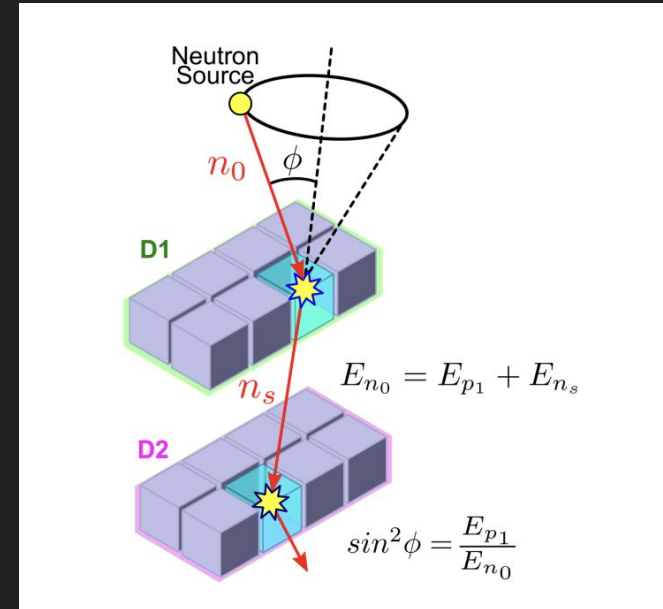
Caden Tebow, Georgia
De Nolfo, Roger
Pujadas, J.G Mitchell

HSD Intern Talk

The nDURE instrument simulated in GEANT with a neutron beam (yellow tracks), with scattering and de-excitation gammas (green tracks)

Neutron Spectrometers - nDURE

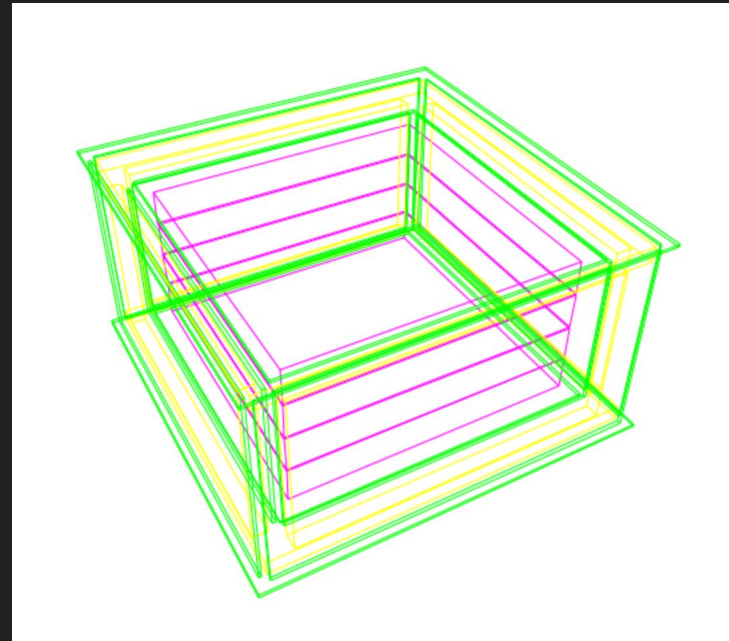
- Neutron Dosimeter for Unknown Radiation Environments → nDURE
- Detect fast albedo neutrons
 - Coming off of the surface of the moon and mars + space, harmful radiation
- Double Scatter Spectrometer
- Energy resolution: At most 30% at 100 MeV
- Efficiency: 0.1% at 10 MeV



[1] Mitchell et Al

Neutron Spectrometers - Single-Scatter

- Simulated using only one of nDURE's scintillators
- Slower neutrons can multi-scatter and deposit most of their energy
- Doesn't rely on low-probability double-scatter: efficiency of 12% at 10 MeV
- We don't care about TOF component, meaning we do not capture the full energy of most neutrons

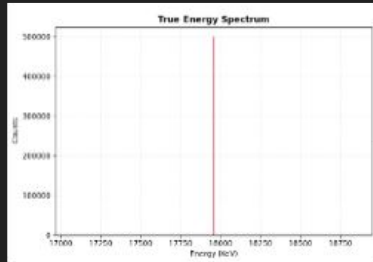


Simulation Model for Single-Scatter Detector
Pink: Scintillator Layers (10x10x5cm)
Yellow: ACD Layers
Green: Aluminum Layers

GEANT Simulation Pipeline

Detector
Effects

Truth: Unknown outside of simulation



GEANT-4
Simulation

Obtain hits in
sensitive volumes

E_{TOF}

Reconstruct Kinetic
Energy: E_{tof}

Smear neutron time
of flight based on
timing resolution

$$\hat{E} = \hat{E}_p + \hat{E}_{TOF}$$

Obtain smeared,
reconstructed
energies

E_p

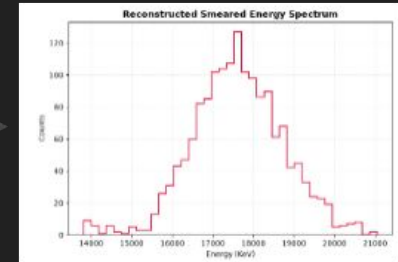
Reconstruct Recoil
Proton Energy: E_p

Smear Recoil
Proton Energy using
Poisson Statistics

Double Scatter

Single Scatter

Reconstruction: What detector
actually “sees”



- Generate truth distribution
- Run it through simulation pipeline
- Recover reconstructed distribution

$$N_{pe} = E_p \cdot conversion \cdot \eta_{trans} \cdot PDE$$

$$\hat{N}_{pe} = Poisson(N_{pe})$$



Simulation Limitations - Inelastic Scattering and Spallation

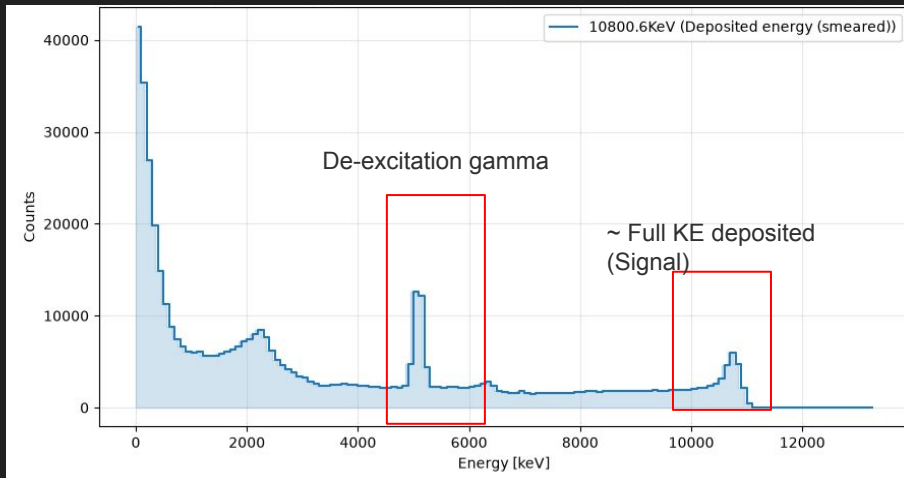
- Simulation: Record exact energy deposited by neutron
- Real life: Record SiPM current proportional to number of optical photons produced
- “Bad” interactions: lose energy to heat, create less optical photons, or break kinematic assumption
 - Spallation: $n + \text{c12} \rightarrow n' + \text{alphas, t, d, etc}$
 - Inelastic Scatter: $n + \text{c12} \rightarrow n' + \text{c12}^*$
 - Capture: $n + \text{c12} \rightarrow n' + \text{c13}$
- My simulations show these **would** become non-negligible around 100 MeV for the single-scatter device
 - However, both detectors have a system that vetoes these bad events at a rate that accounts for their effects

Assumed Alpha Quench	Electron-Equivalent	Photons	How Much 27,000 Photons / MeV Overestimates
0.08	1.446 MeVee	39,039	9.1x
0.10	1.702 MeVee	45,954	7.8x
0.15	2.342 MeVee	63,241	5.6x

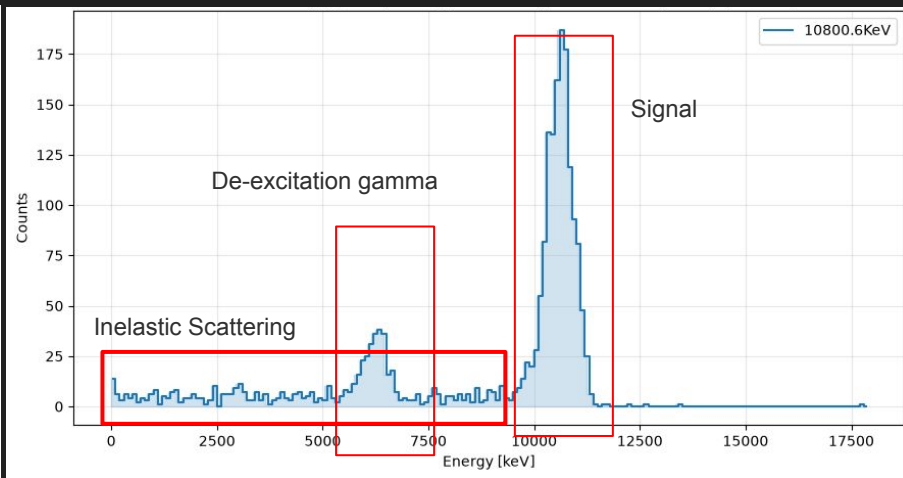
Comparing Single and Double-Scatter Detectors

- Can a Single-Scatter detector be as effective as nDURE at higher energies?
- Three characteristics to compare:
 - Energy resolution (accuracy)
 - Energy orthogonality (ability to distinguish between different energies)
 - “Unfoldability” (effectiveness in recovering the true neutron energy spectra)

Single-Scatter Reconstructed Spectra at 10.8 MeV



nDURE Reconstructed Spectra at 10.8 MeV



Simulation Statistics - nDURE

- For 1 MeV to 1 GeV, Simulated 500,000 events per energy, 69 energies log-scale

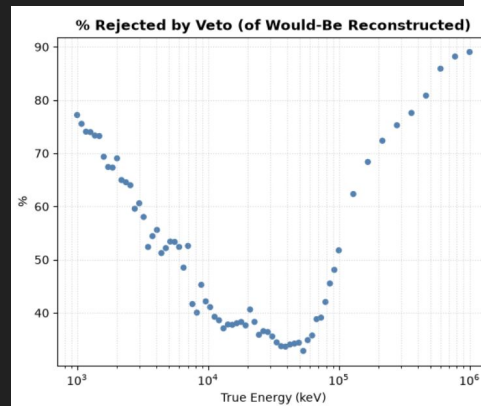
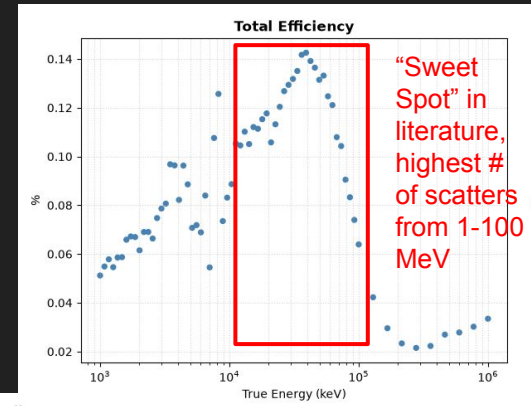
Efficiency:

N Reconstructed:

- | | |
|------------------|-----------------|
| - 1 MeV: 0.05% | - 1 MeV: 500 |
| - 100 MeV: 0.04% | - 100 MeV: 2000 |
| - 1 GeV: 0.035% | - 1 GeV: 1500 |

Why does efficiency rise and then fall?

- Two competing factors:
 - Scintillator Interaction Probability
 - “Reconstruction cross-section”
 - Low E: wider-angle scatters
 - Higher E: more forward scatters, higher free mean path
 - Veto contamination is U shaped



Simulation Statistics - Single-Scatter

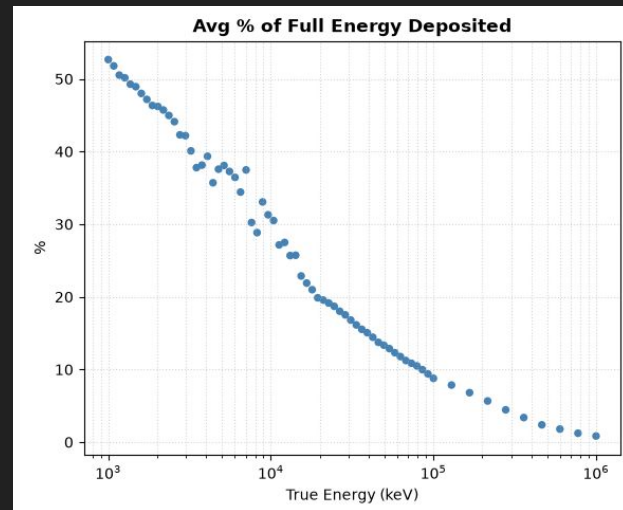
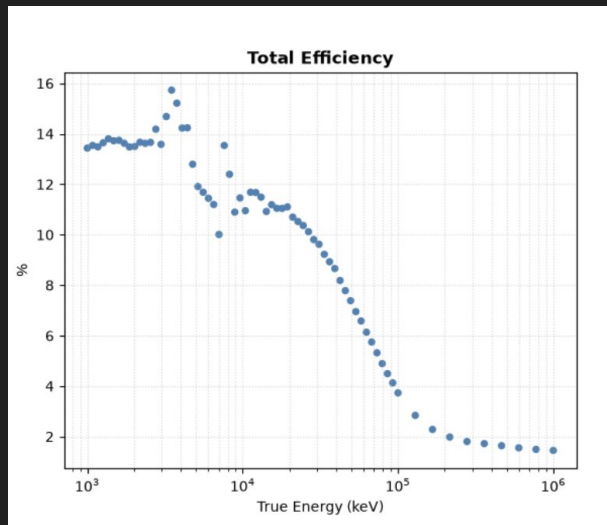
For 1 MeV to 1 GeV, Simulated 100,000 events per energy, 69 energies log-scale

Efficiency:

- 1 MeV: 14%
- 100 MeV: 4%
- 1 GeV: 1%

N Reconstructed:

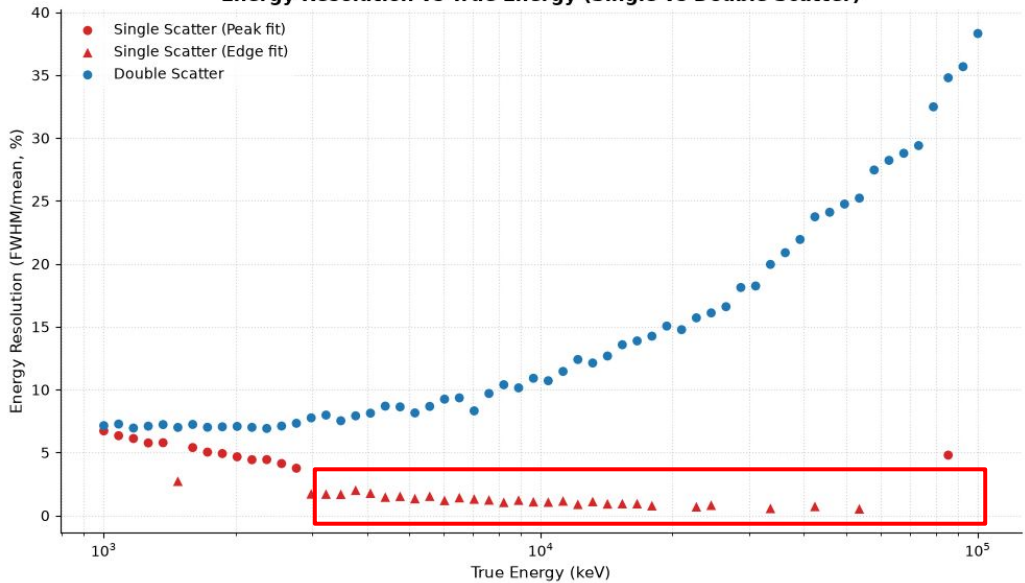
- 1 MeV: 35,000
- 100 MeV: 50,000
- 1 GeV: 25,000



No double-scatter reconstruction, so efficiency looks much more as expected (scintillator cross section lower at higher E)

nDURE and Single-Scatter Energy Resolution

Energy Resolution vs True Energy (Single vs Double Scatter)



Red Triangle: No distinct signal present,
hard to distinguish different energies

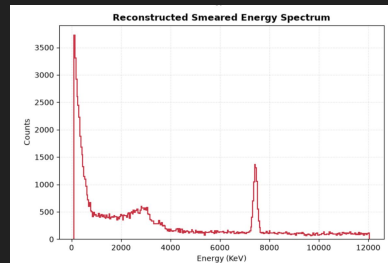
nDURE Energy Resolution:

- 1 MeV: 5%
- 10 MeV: 11%
- 100 MeV: 30%

Single-Scatter appears to outperform
nDURE???

- Highly idealized simulation using monoenergetic beams
- In real life: many different energies at once

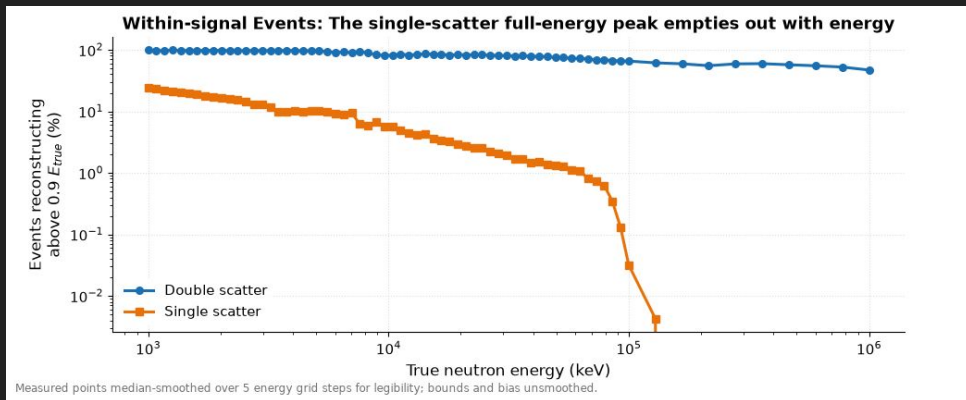
13 MeV
Single-Scatter
Spectra



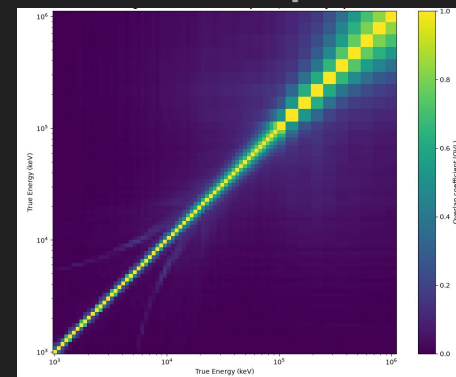
What sets nDURE apart: Energy Orthogonality

- Energy Orthogonality:
 - For two true energies measured on our detector, what percent do their reconstructed distributions overlap by?
- Single-Scatter has poor energy orthogonality despite its edge sharpness

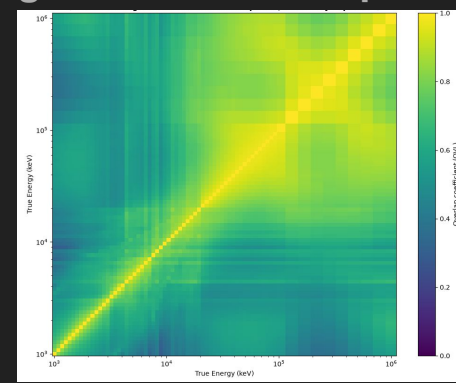
Another Metric: How many events are in the signal?



nDURE Overlap Matrix



Single-Scatter Overlap Matrix

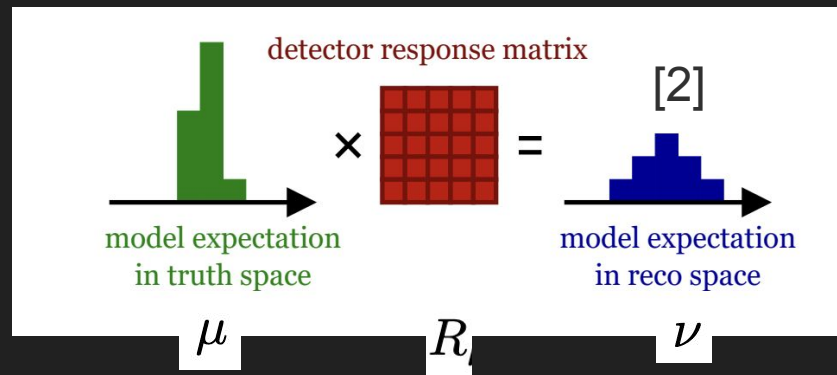


How to Recover True Energy Spectra: Unfolding

Premise: Reconstructed energy spectra are different from their true distributions due to detector/physics/background effects

We model the effects our detector has on the data using a **response matrix R**

Unfolding: Given a response matrix that encodes how our detector “smears” the true distribution, and our reconstructed data, can we recover the true distribution?



Matrix formulation of unfolding problem: We want μ , and we know R and ν . Goal is to find a way to “invert” R in a way that makes physical sense.

$$R\mu = \nu$$

$$\mu = R^{-1}\nu$$

$$\mu \in \mathbb{R}^n$$

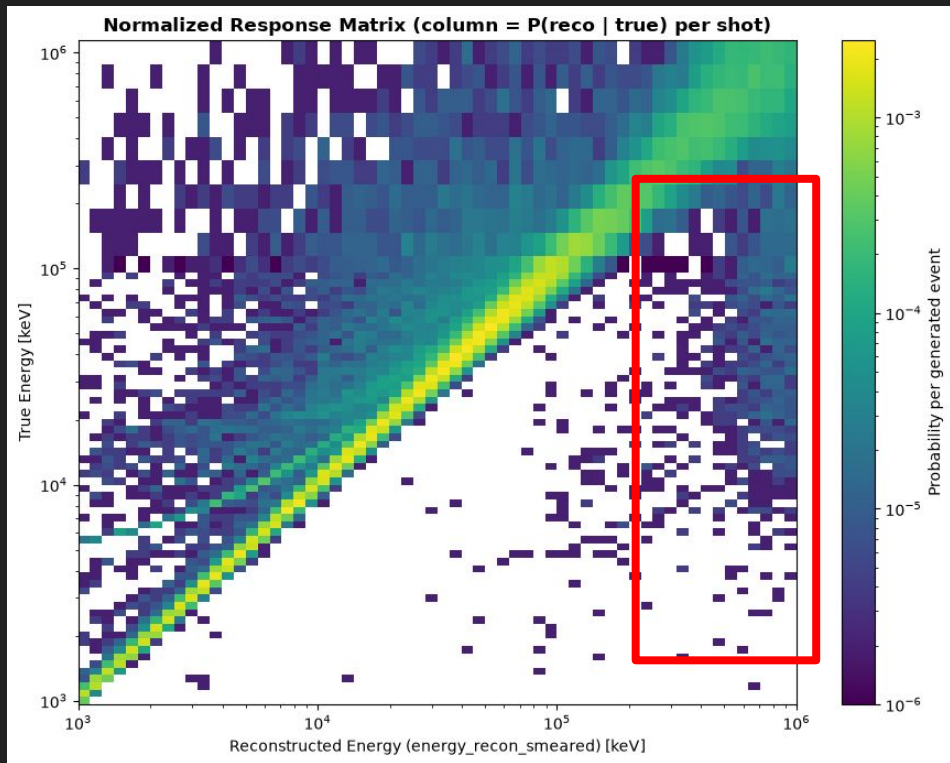
$$\nu \in \mathbb{R}^m$$

Another way to think of response matrix: an operator that tells us how many events end up in reconstruction bin i given a true distribution μ .

$$\nu_i = \sum_j P(i|j)\mu_j$$

$$\nu_i = R_{ij}\mu_j$$

Simulated Response Matrices - nDURE (Double-Scatter)



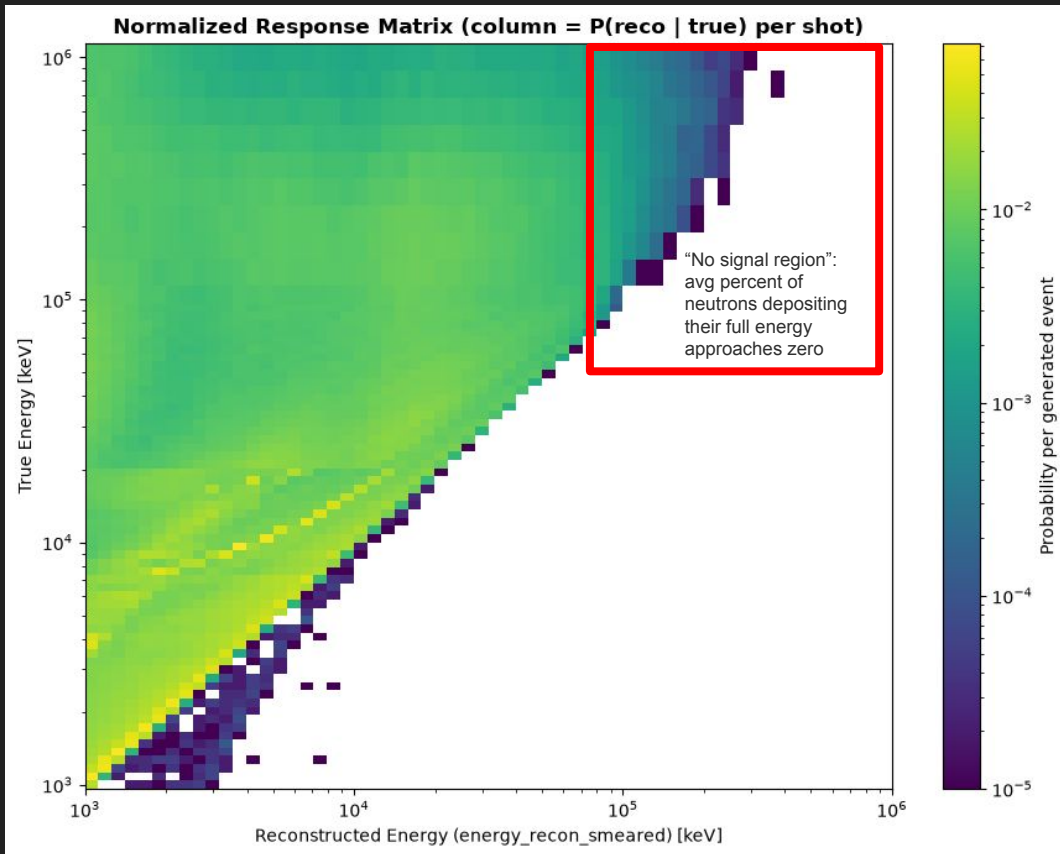
Red Square: This looks wrong...

- A 10 MeV neutron should not be reconstructed as 1 GeV!!!!

What's happening:

- Neutrons inelastically scatter with or are captured by ^{12}C nuclei
- ^{12}C de-excites $\rightarrow$ gamma is released
- Gamma compton scatters in another detector layer $\rightarrow$ we mis-reconstruct it as a neutron
- Gammas travel at speed of light $\rightarrow$ tiny TOF $\rightarrow$ massive reconstructed energy
- We can fix this almost completely with pulse shape discrimination

Simulated Response Matrix - Single-Scatter

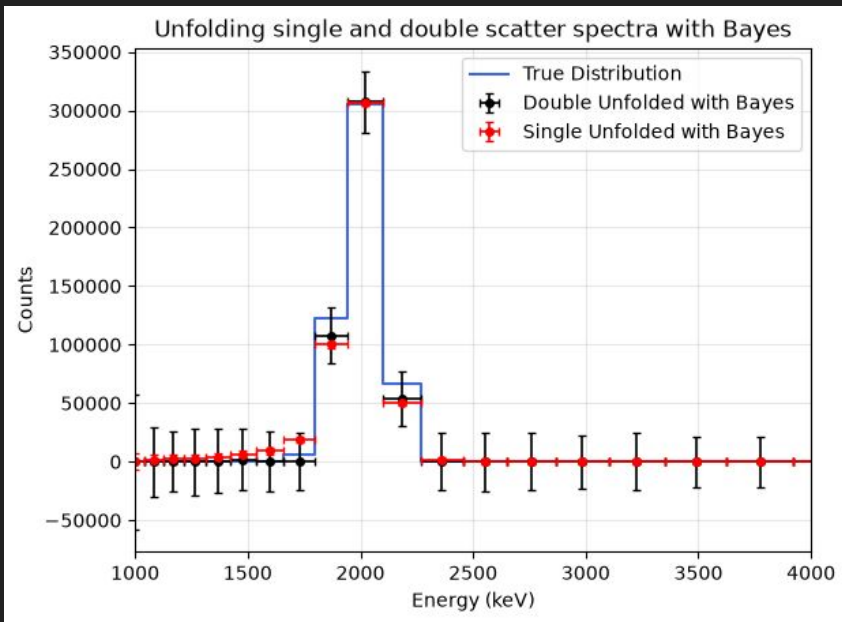


- As true energy becomes greater than 5-10 MeV, loses diagonal structure
- Example: A 10 MeV neutron has a fairly equal chance of being reconstructed at 1 MeV or 10 MeV

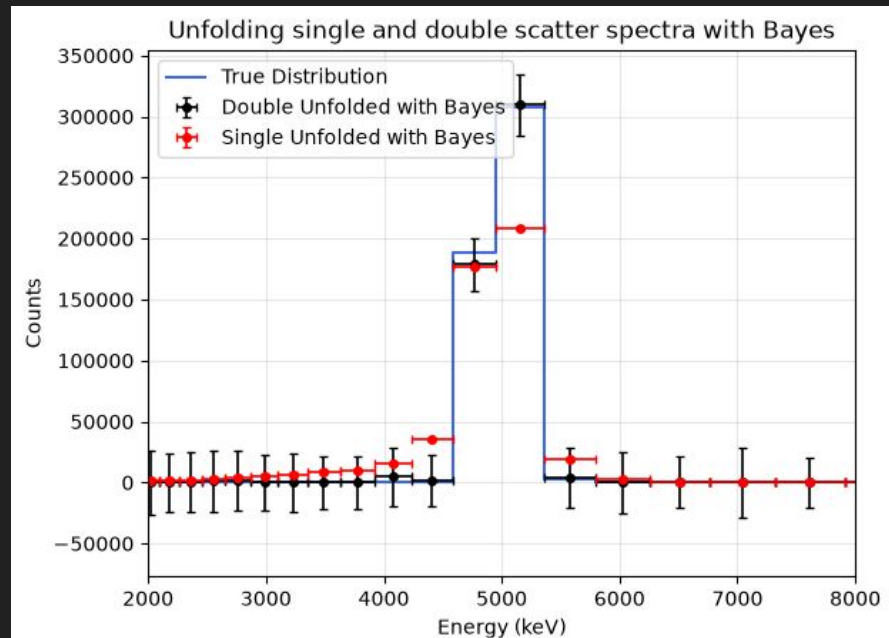
(100,000 events per energy, 69 energies log scale from 1MeV - 1 GeV)

Unfolding Comparison: 2 MeV, 5 MeV Gaussians

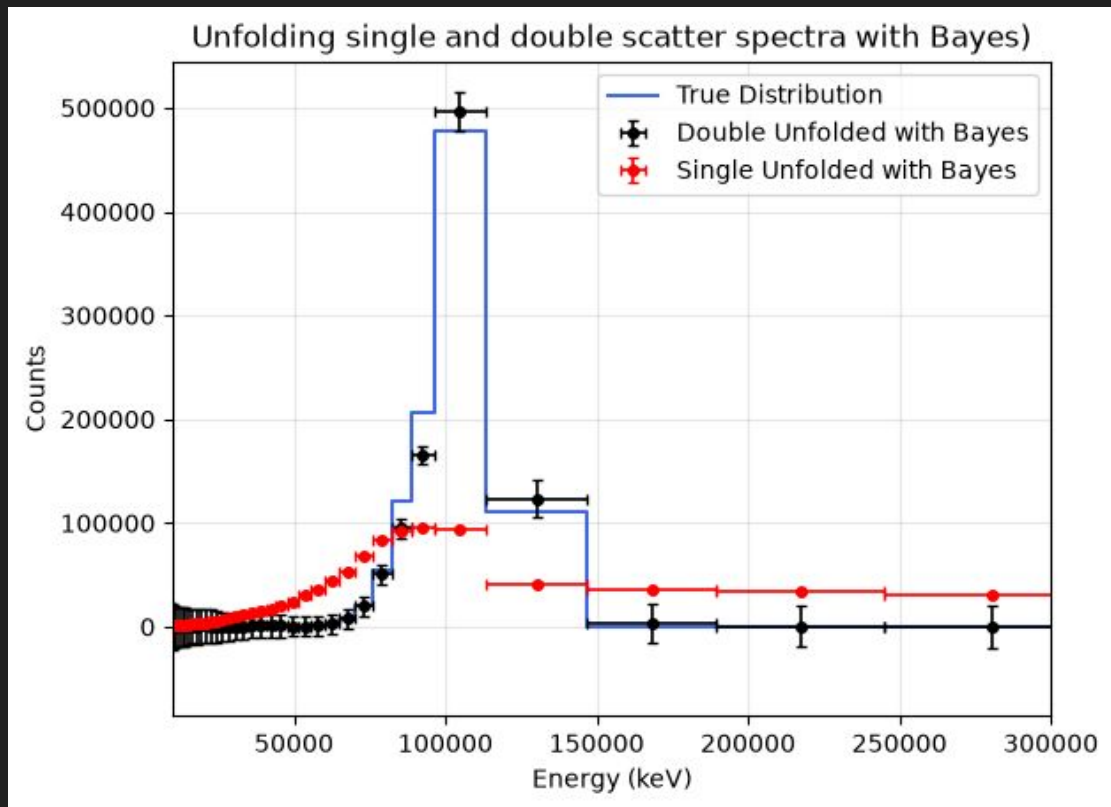
2 MeV Gaussian Input
(true) Spectra



5 MeV Gaussian Input
(true) Spectra



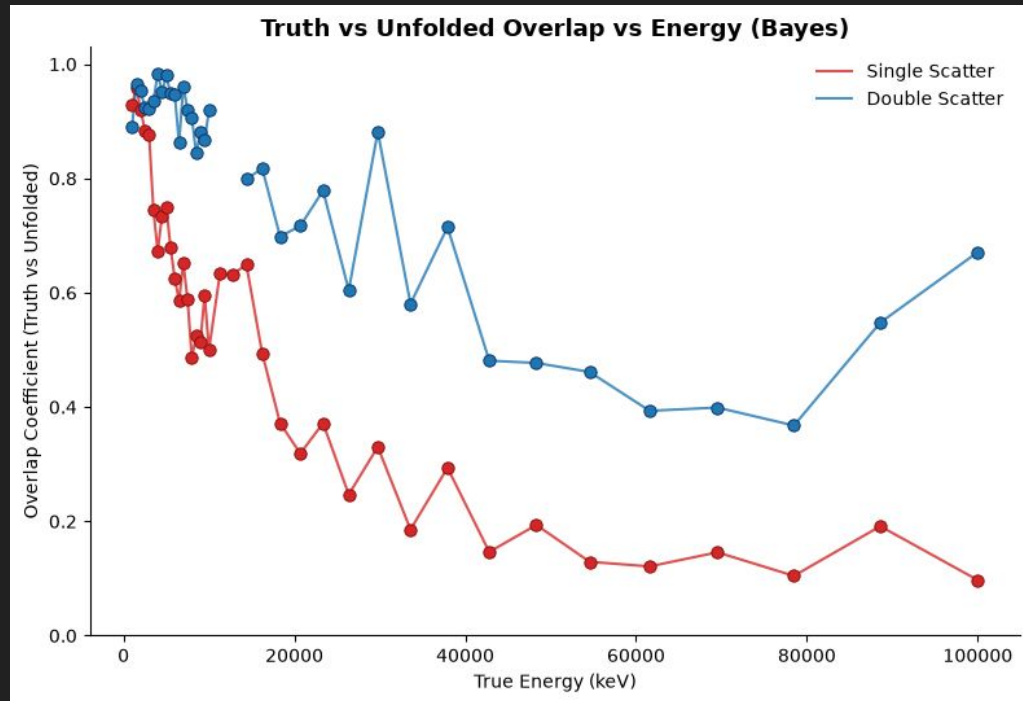
Unfolding Comparison - 100 MeV Gaussian



Unfolding Trends - nDURE Dominates at Higher E

- These trends continue for unfolding gaussians at higher energies using IBU
- Can think of each point as similar to a discrete inner product between truth and unfolded truth histograms
- Actual formula:

$$OVL = \sum_i \min(\mu_i, \hat{\mu}_i)$$





Closing Comments

- Single scatter instruments play a valuable role for situations where a more sophisticated instrument is not possible or needed.
- However, when comparing these specific instruments, if we want good energy resolution + separation + “unfoldability” at higher energies, nDURE is the clear and only choice
- Future investigation will attempt to generalize this analysis:
 - Using different true distributions
 - Using real experimental data in the unfolding pipeline
 - Modifying single-scatter to further examine relationship between scintillating volume and energy resolution



References - Questions?

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